

Formula for Primes

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We've always been told that there is no formula for finding the n th prime number, but as it turns out, there is! The formula was created by C.P. Willans, of which not much is known, back in 1964.

$$p(n) = 1 + \sum_{i=1}^{2^n} \left\lfloor \left(\frac{n}{\sum_{j=1}^i \left\lfloor \left(\cos \pi \frac{(j-1)!+1}{j} \right)^2 \right\rfloor} \right)^{1/n} \right\rfloor$$

The formula can be broken down into pieces, which will make it much easier to understand. Let's start with the innermost piece.

$$\frac{(j-1)!+1}{j}$$

Wilson's theorem states that for this expression, every time j is not a prime number and not 1, it will return a non-integer. Willans then used the cosine pi part in order to make our prime detector output 0s and 1s instead of integers and non-integers. The function $f(x) = \cos(\pi x)$ returns 1 every time x is even and returns -1 every time x is odd, and whenever x is a non-integer, it returns a value between 1 and -1. This means that the cosine part of Willans' formula returns 1 or -1 whenever j is prime or 1 and returns a number between 1 and -1 whenever j is composite. Then, by squaring the cosine expression and adding a floor function around it, Willans makes it so that when j is prime or 1, it returns 1, and returns 0 if j is composite. Now let's look at the whole denominator.

$$\sum_{j=1}^i \left\lfloor \left(\cos \pi \frac{(j-1)!+1}{j} \right)^2 \right\rfloor$$

The summation from $j = 1$ to i basically makes the denominator of the whole fraction return 1 plus the amount of primes up to i . But where do we go from here? How does counting primes help us find the n th prime? Let k represent what the denominator outputs. What Willans did was find a way to mathematically express the question, is the number of primes up to 1 less than k ? Is the number of primes up to 2 less than k ? Is the number of primes up to 3 less than k ? And so on until the answer is no. The moment the answer is no, it will be when we've hit the n th prime. Now that we have a way to count the number of primes up to any integer i , the next challenge is to determine when i reaches the n th prime number. Let's expand our views of the function to see the next step that Willans takes.

$$\left\lfloor \left(\frac{n}{\text{PrimeCount}(i) + 1} \right)^{1/n} \right\rfloor$$

Let's understand this part by setting

$$Q = \frac{n}{\text{PrimeCount}(i) + 1}$$

1. Compute the Quotient:

- **When i is less than the n th prime:**

- The number of primes up to i is less than n .
- Therefore, $\text{PrimeCount}(i) + 1 \leq n$.
- This makes $Q \geq 1$.

- **When i reaches or exceeds the n th prime:**

- The number of primes up to i is at least n .
- Therefore, $\text{PrimeCount}(i) + 1 > n$.
- This makes $Q < 1$.

2. Raise the Quotient to the power of $1/n$:

$$R = Q^{\frac{1}{n}}$$

- **When $Q \geq 1$:**

- $R \geq 1$ because any number greater than or equal to 1 raised to a positive power yields a result greater than or equal to 1.

- $R < 1$ because a number less than 1 raised to a positive power yields a result less than 1.

3. Apply the Floor Function

$$\lfloor R \rfloor$$

- **When $R \geq 1$:**

- The floor of R is 1

- **When $R < 1$:**

- The floor of R is 0

But since $R \geq 1$ only when $Q \geq 1$ (i is less than the n th prime) and $R < 1$ when $Q < 1$, the floor function effectively outputs 1 when i is less than the n th prime and 0 when i is greater than or equal to the n th prime.

Now lets interpret the final part of Willans' formula.

$$\sum_{i=1}^{2^n} \left\lfloor \left(\frac{n}{\sum_{j=1}^i \left\lfloor \left(\cos \pi \frac{(j-1)!+1}{j} \right)^2 \right\rfloor} \right)^{1/n} \right\rfloor$$

Why sum to 2^n though? This is because we need to sum over a range of i values that is guaranteed to include the n th prime. Bertrand's Postulate tells us that there's always at least one prime between m and $2m$ for $m \geq 2$. By extending i up to 2^n , we ensure that the n th prime is within our range.

A quick deviation: Proof of Bertrand's Postulate

One of the simplest proofs of Bertrand's Postulate was given by Paul Erdős in 1932. He begins by considering the binomial coefficient given by:

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2}$$

Using Sterling's approximation for factorials, $n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$, we can estimate the size of $\binom{2n}{n}$ as $\frac{4^n}{\sqrt{\pi n}}$. Because of the 4^n , we can conclude that $\binom{2n}{n}$ grows exponentially. The goal is to understand how prime numbers contribute to this value. Erdős specifically focuses on the prime factorization of $\binom{2n}{n}$. $(2n)!$ is the product of all the integers from 1 to $2n$ while $(n!)^2$ is the product of all integers from 1 to n , repeated twice. When we divide $(2n)!$ by $(n!)^2$, a bunch of numbers cancel out and what's left can be analyzed to understand the contribution of prime numbers. We want to see how often each prime p divides $\binom{2n}{n}$. For a given prime p , the exponent e_p in $(2n)!$ is given by:

$$e_p = \left\lfloor \frac{2n}{p} \right\rfloor + \left\lfloor \frac{2n}{p^2} \right\rfloor + \left\lfloor \frac{2n}{p^3} \right\rfloor + \dots$$

The same formula applies to $(n!)^2$ and we can subtract the latter from the former to get the exponent of p in $\binom{2n}{n}$. Erdős then does proof by contradiction. Let there be no primes in $(n, 2n)$, then every prime number that divides $\binom{2n}{n}$ must be $\leq n$. From here, we can express $\binom{2n}{n}$ as a product of primes.

$$\binom{2n}{n} = \prod_{p \leq n} p^{e_p}$$

where e_p is exponent indicating how many times each prime p divides $\binom{2n}{n}$. Recall that the binomial coefficient could be estimated by $\frac{4^n}{\sqrt{\pi n}}$. The product $\prod_{p \leq n} p^{e_p}$ cannot grow as fast as the approximation of the binomial coefficient. The Prime Number Theorem (the proof of which is too complex to include in this paper) states that the amount of primes less than a given number n can be estimated as $\frac{n}{\ln(n)}$ which does not grow nearly as rapidly as the exponential function that approximates $\binom{2n}{n}$. Here lies the contradiction. $\binom{2n}{n}$ cannot be as large as $\frac{4^n}{\sqrt{\pi n}}$ if it only has prime factors up to n . Therefore, by proof of contradiction, there must exist at least 1 prime in the interval $(n, 2n)$, which concludes Erdős' proof of Bertrand's postulate. And now back to the main proof...

Interpretation:

- For each i less than the n th prime, the expression outputs 1.
- For i equal to or greater than the n th prime, it outputs 0.
- Thus, the sum counts the number of integers i less than the n th prime.

This means that the total sum is $p(n) - 1$ where $p(n)$ is the n th prime number. By adding 1 to the total sum, we get $p(n)$. However, this formula is less than efficient. I've created a simple C++ script to calculate the n th prime using this formula.

```
#include <iostream>
#include <math>
```

```

#include <vector>
using namespace std;

int prime(int n) {
    int result = 1;
    for (int i = 1; i <= pow(2, n); ++i) {
        double inner_sum = 0;
        for (int j = 1; j <= i; ++j) {
            inner_sum += floor(pow(cos(M_PI * (tgamma(j) + 1) / j), 2));
        }
        result += floor(pow(n / inner_sum, 1.0 / n));
    }
    return result;
}

int main() {
    int n = 10;
    cout << "Result for prime(" << n << "): " << prime(n) << endl;
    return 0;
}

```

Using this script to calculate the 10th prime, you get -2147483648. Obviously, this is wrong. The reason for this erroneous output is because the int type in C++ usually has a limit of 2147483647. Any calculations exceeding this limit would wrap around and produce incorrect results. If just trying to calculate the 10th prime is this tough to do using a computer, imagine trying to calculate the 1000th prime. Even if the limit in C++ is removed, at the very least a computer still has to run the main loop 2^{1000} times, each time adding another term to the sum. Iterating through 2^{1000} terms takes up an enormous amount of resources, which makes C.P Willans' formula correct but not the most useful. This is probably why even though there is a formula for finding the nth prime, most people thought there wasn't.